

- 1) What is unsupervised learning? What are the Common approaches and Challenges.

Unsupervised Learning

Definition

- A machine learning technique that uses **unlabeled data** (no predefined output).
- The model **finds hidden patterns and structures** without supervision.
- Goal: **Group data based on similarities** and represent it in a simplified format.

How it Works (Simple Flow)

1. **Input:** Unlabeled data given to model.
2. **Process:** Model searches for similarities/patterns.
3. **Algorithm:** Applies clustering (K-means, Hierarchical) or association (Apriori).
4. **Output:** Data grouped into clusters or rules discovered.

Common Approaches

1. **Clustering** – Grouping similar data points together.
 - Example: K-means, Hierarchical, DBSCAN.
2. **Association** – Finding relationships between variables.
 - Example: Market Basket Analysis (bread → butter).

Challenges

1. **Unverifiable** – True number of clusters or structure is unknown.
2. **Hard to Interpret** – Results may be meaningless without domain knowledge.
3. **Need for Supervision** – Human validation often required.
4. **Misalignment with Goals** – Discovered patterns may not match application needs.

- 2) Write a note on Logistic regression.

Logistic Regression (10 Marks)

Definition

- Logistic Regression is a **supervised ML algorithm** used for **classification problems**.
- Predicts **probability of class membership** (0/1, Yes/No) using **Sigmoid function**.

Equation:

$$\hat{y} = \sigma(b_0 + b_1x) = \frac{1}{1 + e^{-(b_0 + b_1x)}}$$

Types of Logistic Regression

1. **Binomial:** Two classes (0/1, Pass/Fail).
 2. **Multinomial:** 3 or more unordered classes (Cat/Dog/Sheep).
 3. **Ordinal:** 3 or more ordered classes (Low/Medium/High).
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Working Principle

- Step 1: Compute linear combination $\rightarrow z = \mathbf{b}_0 + \mathbf{b}_1\mathbf{x}$
 - Step 2: Pass z through sigmoid $\rightarrow$ get probability
 - Step 3: Apply decision threshold ($p > 0.5 \rightarrow$ Class 1, else Class 0)
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Key Assumptions

- ✓ **Binary Response Variable** – Target must be categorical
 - ✓ **Independent Observations** – No relation between samples
 - ✓ **No Multicollinearity** – Predictors not highly correlated
 - ✓ **No Extreme Outliers** – Avoid data distortion
 - ✓ **Linearity in the Logit** – Predictors have linear relation with log-odds
 - ✓ **Sufficient Sample Size** – Enough data points per variable
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Applications

- Spam detection
- Medical diagnosis
- Risk prediction (loan default, churn analysis)

3) List and explain Ensemble Techniques

Ensemble Techniques

Ensemble Learning combines multiple models to improve performance and accuracy. It reduces errors and increases the robustness of predictions.

1. Max Voting

- Used for **classification problems**.
 - Each model gives a prediction (vote).
 - The class with the **majority votes** is selected as the final prediction.
 - Example: If 3 models predict Class A and 2 models predict Class B $\rightarrow$ Final class = A.
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2. Averaging

- Used for **regression or probability-based classification**.
 - Takes the **average of predictions** from all models.
 - Example: Predictions = [5, 4, 5, 4, 4] → Final result = Average = 4.4
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3. Weighted Average

- Similar to averaging, but **weights are assigned** to models based on importance.
 - More reliable models get **higher weights**.
 - Example: Final prediction = $(w_1 \times \text{pred}_1 + w_2 \times \text{pred}_2 + \dots) / (\text{sum of weights})$.
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4. Stacking (Advanced Technique)

- Predictions of **multiple base models** (e.g., Decision Tree, KNN, SVM) are used as input to a **meta-model** (e.g., Logistic Regression).
 - The meta-model learns from base model outputs and gives the final prediction.
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5. Blending (Advanced Technique)

- Similar to stacking but uses a **holdout validation set** instead of cross-validation.
 - Base models are trained on training data, predictions are made on validation set → final model is built on these predictions.
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6. Bagging (Bootstrap Aggregating)

- Creates **multiple subsets** of data using **sampling with replacement**.
 - A model is trained on each subset independently.
 - Final prediction = **majority voting (classification)** or **average (regression)**.
 - Reduces variance and prevents overfitting.
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7. Boosting

- Sequential process where each model **corrects errors** of the previous one.
- Misclassified samples get **higher weights** in the next model.
- Final prediction is a **weighted combination** of all models.
- Improves accuracy but may overfit if not controlled.